

[ACJC/2025/Q2]

2 Name your Jupyter Notebook as:

TASK2_<your name>_<centre number>_<index number>.ipynb

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 2.1
        Program code
        Output:
```

Task 2.1

Write a function `quicksort(lis)` that sorts a list of positive integers in ascending order, using a quicksort algorithm that works in place. [8]

Write a function `insertionsort(lis)` that sorts a list of positive integers in ascending order, using an insertion sort algorithm that works in place. [3]

Task 2.2

Generate the following four lists:

- `lis1` and `lis2` are a list of the integers from 1 to 1000 inclusive, in ascending order;
- `lis3` is a list of the integers from 1 to 1000 inclusive, in random order;
- `lis4` is a copy of `lis3`.

The `timeit` library provides a way to time how long a snippet of code takes to run. For example, the following code block prints the time it takes for `quicksort(lis1)` to run.

```
import timeit
setup_code = "from __main__ import quicksort, lis1"
time = timeit.timeit("quicksort(lis1)", setup_code, number=1)
print(time)
```

Output the time it takes to run

- `quicksort(lis1)`
- `insertionsort(lis2)`
- `quicksort(lis3)`
- `insertionsort(lis4)`

[4]

Save your Jupyter Notebook for Task 2.